THE LAWS OF SOFTWARE TEAMS

Hyrum's law

Every observable behavior of your API will be depended on by somebody

swipe →

Ivan Rykov

01 THE HOOK

You fixed a typo
You broke a customer

CHANGELOG — fix: typo in an error message

```
- "Invaild token"
+ "Invalid token"
```

A CUSTOMER'S INTEGRATION — DOWN

```
if err.message == "Invaild token":
    refresh_session()
# now never true
```

Their code read the exact message

illustration

02 THE LAW

Hyrum's law: implicit interfaces

Hyrum Wright · software engineer at Google

> "With a sufficient number of users of an API, it does not matter what you promise in the contract: **all observable behaviors of your system will be depended on by somebody**"

hyrumslaw.com · named "Hyrum's law" by Titus Winters
also in Software Engineering at Google (O'Reilly, 2020)

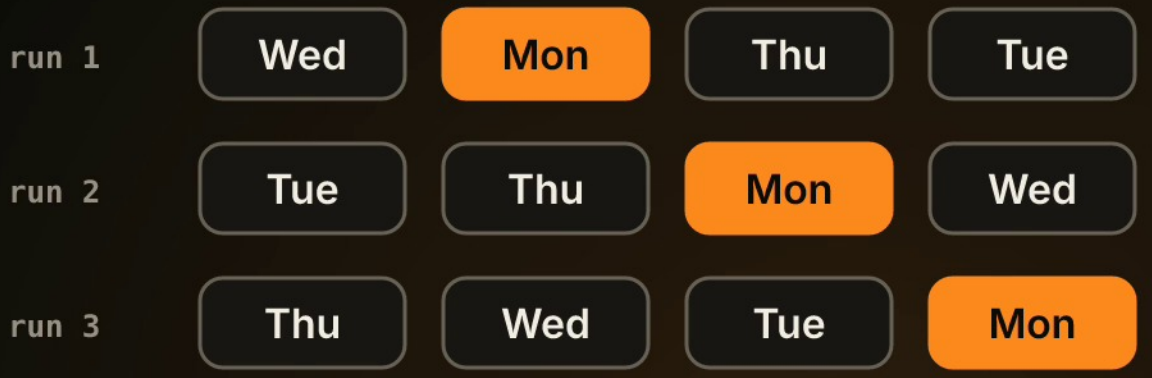
The contract is small
The interface is not



illustration · everything users can see becomes the interface

Go made map order random, on purpose

```
for day := range week { … }
```



A TEST THAT EXPECTS “Mon” FIRST

Before: it passed on one machine and failed on another

Go 1: it fails at once, and gets fixed early

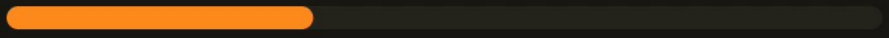
Go 1 release notes, March 2012 · orders shown are illustrative

A bug fix can break a user

UPDATE NOTES

Fix: holding the spacebar no longer overheats the CPU

CPU heat



USER FEEDBACK

"You broke my workflow"

He used the CPU heat as a keyboard shortcut

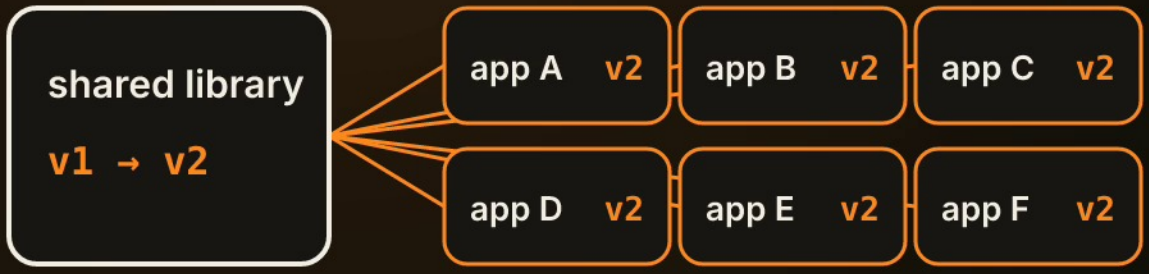
Every behaviour has a user

xkcd 1172 "Workflow" by Randall Munroe · described in words

Who changes it, migrates it

THE CHURN RULE · GOOGLE · 2012

"Infrastructure teams must do the work to move their internal users to new versions themselves or do the update in place"



The team that changes the library moves every user

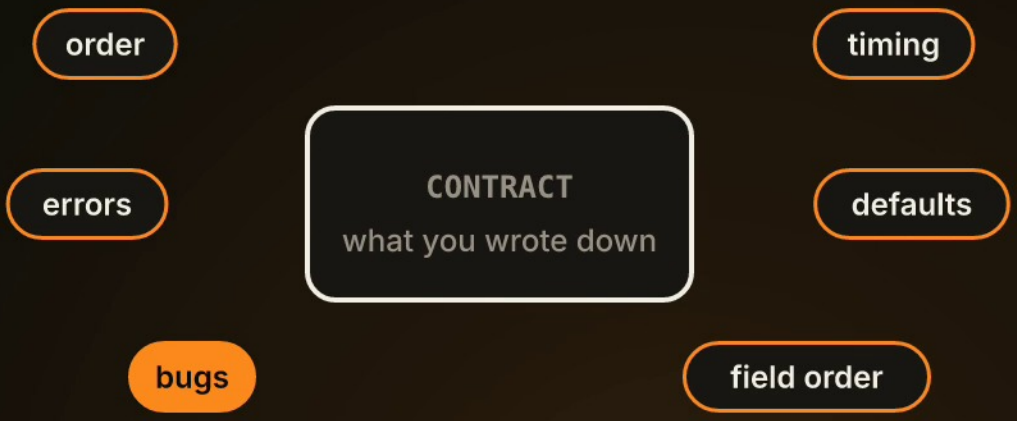
Software Engineering at Google, ch. 1 (O'Reilly, 2020) · diagram illustrative

07 WHAT TO DO

Treat every change as a contract change

1. **Version the API**
   old and new versions run side by side
2. **Announce deprecations early**
   dates, warnings and a migration guide
3. **Test what users depend on**
   consumer-driven contract tests
4. **Randomise what you never promised**
   the Go trick: order, IDs, timing

Your real API is what people use,



not what you wrote down

TAKEAWAY

Your real API is what people use,
not what you wrote down

Sources: Hyrum Wright, hyrumslaw.com · Software Engineering at Google, 2020 · Go 1 release notes, 2012

next: Kernighan's law